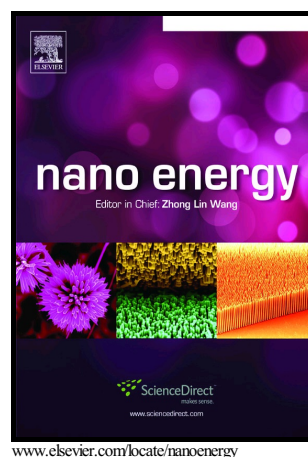


Achieving commercial-level mass loading in ternary-doped holey graphene hydrogel electrodes for ultrahigh energy density supercapacitors

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**Achieving commercial-level mass loading in ternary-doped holey
graphene hydrogel electrodes for ultrahigh energy density
supercapacitors**

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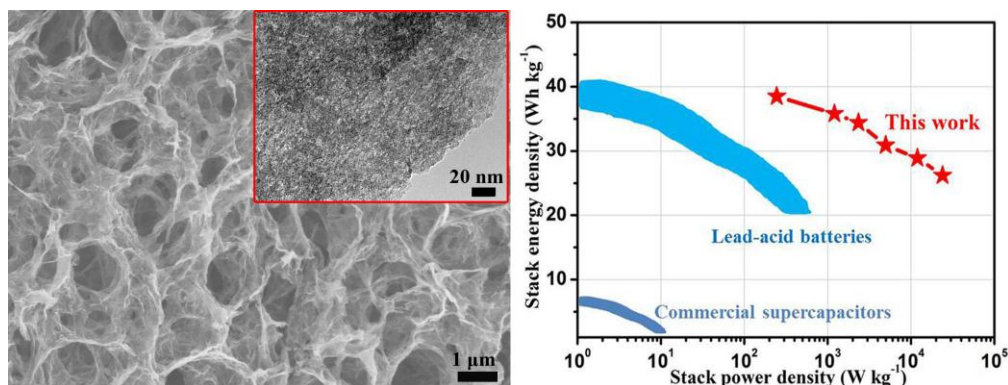
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Abstract

Enabling fast ion diffusion in thick electrodes ($100\text{--}200\text{ }\mu\text{m}$, $\sim 10\text{ mg cm}^{-2}$) is critical for their practical application in state-of-the-art supercapacitors (SCs). We developed a three-dimensional (3D) boron, nitrogen, and phosphorus ternary-doped holey graphene hydrogel (BNP-HGH) film to achieve an optimized porous structure with a high electrical conductivity, large ion accessible surface area, efficient electron and ion transport pathways, as well as high ion adsorption capacity. The binder-free BNP-HGH electrode can deliver a specific capacitance of 350 F g^{-1} and a volumetric capacity of 234 F cm^{-3} , which are the best performance reported so far for graphene-based SCs using an organic electrolyte. Fully packaged SCs using the BNP-HGH electrodes with a commercial level graphene mass loading ($150\text{ }\mu\text{m}$, $\sim 10\text{ mg cm}^{-2}$) can deliver ultrahigh stack gravimetric and volumetric energy densities of 38.5 Wh kg^{-1} and 57.4 Wh L^{-1} , respectively, which are comparable to those of lead-acid batteries ($35\text{--}40\text{ Wh kg}^{-1}$ and $50\text{--}90\text{ Wh L}^{-1}$) while maintaining an ultrahigh power density of 83 kW kg^{-1} ($\sim 55\text{ kW L}^{-1}$) as well as a long cycle life (81.3 % capacitance retention over 50,000 cycles). The high energy and power densities bridge the gap between traditional SCs and batteries, and should be very useful in practical applications.

Graphical abstract



Keywords: supercapacitors, B, N, P, ternary-doped, holey graphene hydrogel, commercial level graphene mass loading, ultrahigh energy density

1. Introduction

Supercapacitors (SCs) have been regarded as one of most promising energy storage devices due to the fast charging capability and long cycle life.¹⁻⁴ Unfortunately, the relatively poor energy density, at least one order of magnitude lower than those of traditional batteries limits the broad application of these SCs.⁵⁻⁶ Although some recent works have claimed to enhance the energy density of SCs to 100-150 Wh kg⁻¹ (against the mass of the active materials in the electrodes), almost comparable to those of lithium ion batteries (120-170 Wh kg⁻¹),⁷⁻¹⁵ the thickness and active materials mass loading of these SCs electrodes are only 5-15 μm and 0.5-1 mg cm⁻², respectively, which are much lower than the values required in commercial SCs (100-200 μm , ~10 mg cm⁻²). Thus, it is highly desirable to develop new electrode materials that can deliver a high energy density at high active materials mass loading without sacrificing

the power density and cycle life.

In general, maintaining the energy density in thicker or higher active materials mass loading electrodes requires proportionally efficient ion and electron currents across a longer charge transport distance.⁴ To this end, a high-performance SC electrode needs excellent electrical conductivity, large ion-accessible surface area and efficient ion transport pathway.¹⁷⁻²⁵ For example, the activated microwave expanded graphene oxide (as-MEGO) provides high surface area and efficient paths for ion diffusion, and the energy density of this electrode can be up to 55 Wh kg⁻¹ at a mass loading of 10.4 mg cm⁻².²¹ Noted this electrode includes a binder, which is dead weight on contributing energy density. More recently, it has been shown that the liquid electrolyte-mediated chemically converted graphene (EM-CCG) film with a continuous ion transport network in the whole film is directly used as binder-free electrode (80 μm, ~10 mg cm⁻²) and leads to an energy density of 88 Wh kg⁻¹.²⁶

In this work, we report a three-dimensional (3D) boron, nitrogen, and phosphorus ternary-doped holey graphene hydrogel (BNP-HGH) as a binder-free electrode for high energy density SCs. The boron, nitrogen, and phosphorus doping in carbon networks can not only facilitate charge transfer between neighboring carbon atoms, and thus enhance electric double layer capacitive (EDLC) performance; but also give rise to pseudocapacitance by faradaic reactions between electrode material and electrolyte.²⁷⁻²⁹ Additionally, there are two types of different pores in the electrode, where the nanoholes are formed throughout each graphene sheet, the macropores are generated by stacking the graphene sheets with numerous curls and wrinkles.²² The

macropores can thus be acted as a ‘highway’ for electrolyte transportation (Supporting Information Figure S1).³⁰⁻³² As a consequence, the resulting BNP-HGH electrode can deliver a gravimetric capacitance of 350 F g^{-1} and a volumetric capacitance of 234 F cm^{-3} , which are the highest to date for graphene-based SCs in an organic electrolyte. We further demonstrated that our BNP-HGH electrode with a commercial level graphene mass loading ($\sim 150 \text{ }\mu\text{m}$, $\sim 10 \text{ mg cm}^{-2}$) exhibits ultrahigh gravimetric and volumetric energy densities of 134 Wh kg^{-1} , and 89 Wh L^{-1} , respectively. Normalized by the total weight or volume of the whole SC including two electrodes, two current collectors, electrolyte, one separator and total wrapped parafilm package, our device still delivers an impressive stack gravimetric and volumetric energy densities of 38.5 Wh kg^{-1} and 57.4 Wh L^{-1} , respectively, which are comparable to those of lead-acid batteries ($35\text{--}40 \text{ Wh kg}^{-1}$ and $50\text{--}90 \text{ Wh L}^{-1}$).

2. Experimental Section

2.1 Preparation of Graphene oxide (GO)

GO was first synthesized according to the literature reported by Xu et al. Briefly, natural graphite powder (20 g) was put into an $80 \text{ }^{\circ}\text{C}$ solution of concentrated H_2SO_4 (30 mL), $\text{K}_2\text{S}_2\text{O}_8$ (10 g), and P_2O_5 (10 g), and then kept for 6 h to pre-oxidize the graphite powder. After cooling to room temperature, 3.5 L de-ionized (DI) water was added into the mixture, which was then filtrated, washed and vacuum dried before storage. Subsequently, the dried pre-oxidized graphite was oxidized using a modified Hummers’ method: firstly, the pre-oxidized graphite powder (0.5 g) was put

into concentrated H_2SO_4 (12 mL, 0 °C), followed by gradually adding KMnO_4 (1.5 g) under vigorous stirring. Secondly, the mixture was stirred at 35 °C for 2 h, and then diluted with DI water (24 mL). Successively, the resultant mixture was stirred at an elevated temperature of 80 °C for 0.5 h. After cooling to room temperature, an additional 70 mL DI water was added, and shortly, 1.25 mL 30 % H_2O_2 was added. Lastly, the mixture was washed with 5 wt % HCl aqueous solution (125 mL) to remove metal ions, followed by washing with DI water three times and centrifugation. Dialysis was carried out to further remove the remaining metal species.

2.2 Preparation of boron, nitrogen, and phosphorous ternary doped graphene hydrogel (BNP-GH)

The BNP-GH can be easily prepared by a modified hydrothermal reduction method. Briefly, a 10 mL GO (2 mg mL^{-1}) aqueous dispersion containing 0.05 M boron phosphate (BPO_4) and 0.05 M ammonium tetrafluoroborate (NH_4BF_4) was first treated by sonication for 30 min, and then the stable mixture was sealed in a Teflon-lined autoclave and hydrothermally treated at 180 °C for 12 h. The autoclave was naturally cooled to room temperature, and the as-prepared graphene hydrogel was taken out with a tweezer and immersed in pure water to remove any impurities for the following experiments. For comparison, the BN-GH, NP-GH, N-GH and GH were also prepared using the same procedure with the addition of the appropriate dopants as needs. For example, ammonium tetrafluoroborate (NH_4BF_4), ammonium dihydrogen phosphate ($\text{NH}_4\text{H}_2\text{PO}_4$) and ammonium hydroxide ($\text{NH}_3\text{H}_2\text{O}$) were employed for BN-GH, NP-GH, N-GH, respectively.

2.3 Preparation of holey graphene oxide (HGO) and boron, nitrogen, and phosphorous ternary doped holey graphene hydrogel (BNP-HGH)

HGO was prepared according to the following procedure: Typically, 10 mL 30 % H_2O_2 and 1-2 ml HCl aqueous solution was mixed with 100 mL 2 mg mL^{-1} GO aqueous dispersion and then heated at 100 °C for 4 h under stirring. The suspension was purified by centrifuging and washing the above mixture to remove the residual H_2O_2 and then re-dispersed in water by ultrasonication to produce a homogeneous HGO aqueous dispersion with a concentration of 2 mg mL^{-1} . The BNP-HGH was prepared by the same method as described in preparation of BNP-GH section.

2.4 Fabrication of BNP-GH, BN-GH, NP-GH, N-GH, B-GH, P-GH, GH, HGH and BNP-HGH based symmetric supercapacitors with aqueous electrolyte

Before assembly, the BNP-GH slices (a thickness of 1 mm) which cut from the purified cylindrical BNP-GH were exchanged with the electrolyte solution (1.0 M H_2SO_4) for 12 h to exchange their interior water with electrolyte. Subsequently, the BNP-GH slices solvated with aqueous electrolytes were placed on the Pt foils, and compressed using hydraulic press at a rate of $\sim 0.5 \text{ cm min}^{-1}$ during which the squeezed electrolytes were removed by filter papers. The samples were kept under a 100 MPa pressure for 2 min to form $\sim 15 \text{ }\mu\text{m}$ well-adhered films on the Pt foils. To make the symmetric supercapacitor, two same BNP-GH films (the areal mass loading of graphene in the BNP-GH films was 1 mg cm^{-2}) on separate Pt foils were directly used as electrodes without any other additives or further treatments such as drying and thermal annealing, and separated by an glass fiber membrane soaked with electrolytes.

All the components were assembled into a layered structure and tightly sealed by parafilm for electrochemical measurements. The BN-GH, NP-GH, N-GH, B-GH, P-GH, GH, HGH and BNP-HGH based symmetric supercapacitors with aqueous electrolytes were fabricated using the similar methods.

2.5 Fabrication of BNP-HGH based flexible all-solid-state supercapacitors with H₂SO₄/PVA gel electrolyte

First, the H₂SO₄/PVA gel electrolyte was prepared as follows: 1 g of H₂SO₄ was added into 10 mL of deionized water, and then 1 g of PVA power was added. The whole mixture was heated to 85 °C under stirring until the solution became clear. Second, the as-prepared BNP-HGH films (the areal mass loading of graphene in the BNP-HGH films was 1 mg cm⁻²) were pressed on the gold coated polyimide substrate. In order to assemble the ASCs, the prepared BNP-HGH films electrode were immersed into the H₂SO₄/PVA gel electrolyte for 2 h, and subsequently maintained at 80 °C for overnight to evaporate the excess of water in the electrolyte. Then the ASCs were assembled by pressing these two electrodes with gel electrolyte together under a pressure of ≈ 2 MPa for 5 min, which can facilitate the polymer gel electrolyte penetrating into each electrode and also form one thin separator layer.

2.6 Fabrication of BNP-HGH, BNP-GH, HGH and GH based symmetric supercapacitors with organic electrolyte

For assembly of supercapacitors with organic electrolyte: the BNP-HGH, BNP-GH, HGH and GH slices were first immersed in pure ionic liquid 1-Ethyl-3-methylimidazolium tetrafluoroborate (EMIMBF₄) under vacuum at 100 °C for 24 h

and by vacuum evaporation of the water residual in the electrodes, and then transferred to a 1.0 M EMIMBF₄ in acetonitrile (AN) solution for another 24 h. Then all the BNP-HGH, BNP-GH, HGH and GH based symmetric supercapacitors were fabricated using the same method as described in preparation of BNP-GH based symmetric supercapacitors with aqueous electrolyte. The assembly of SCs with EMIMBF₄/AN electrolyte was done in a glove box filled with Ar. Thicker film electrodes (an areal mass loading of graphene in the electrodes was 5 and 10 mg cm⁻²) in SCs were prepared by compressing thicker BNP-HGH slices with a thickness of 5 mm and 10 mm under the same pressure. To determine the packing density, we calculated the volume by measuring the radius (r) and thickness (h) of the compressed BNP-HGH film ($V = h \times \pi \times r^2$) and directly weighed the net mass of the vacuum dried BNP-HGH. We can thus obtain the packing density by dividing the net mass by the volume of the HGF film.

2.7 Structural characterization and analysis

The microstructure of the sample was characterized using a Quanta 400 FEG SEM and a Tecnai G2 F20 S-Twin TEM. Due to the hydrogel containing water cannot be directly used for SEM characterization, so we prepared the SEM samples by freeze drying. XRD was used to study the crystal structure of the samples with a Bruker D8 diffractometer; XPS characterization was performed on an Axis Ultra DLD X-ray photoelectron spectroscopy. Raman spectra were also used to analyze the structure information of the as-obtained samples with a LabRAM HR evolution Raman spectrometer. The electrical conductivity of the samples was

recorded with a standard four-probe analyzer.

2.8 Electrochemical characterization and analysis

All electrochemical performance of the samples was measured by a CHI 660C electrochemical workstation (CH Instruments Inc.). The electrochemical impedance spectroscopy (EIS) measurements were performed at open circuit potential with a sinusoidal signal over a frequency range from 100 kHz to 10 mHz at an amplitude of 10 mV. The cycle life tests were conducted by galvanostatic charge-discharge measurements.

3. Result and Discussions

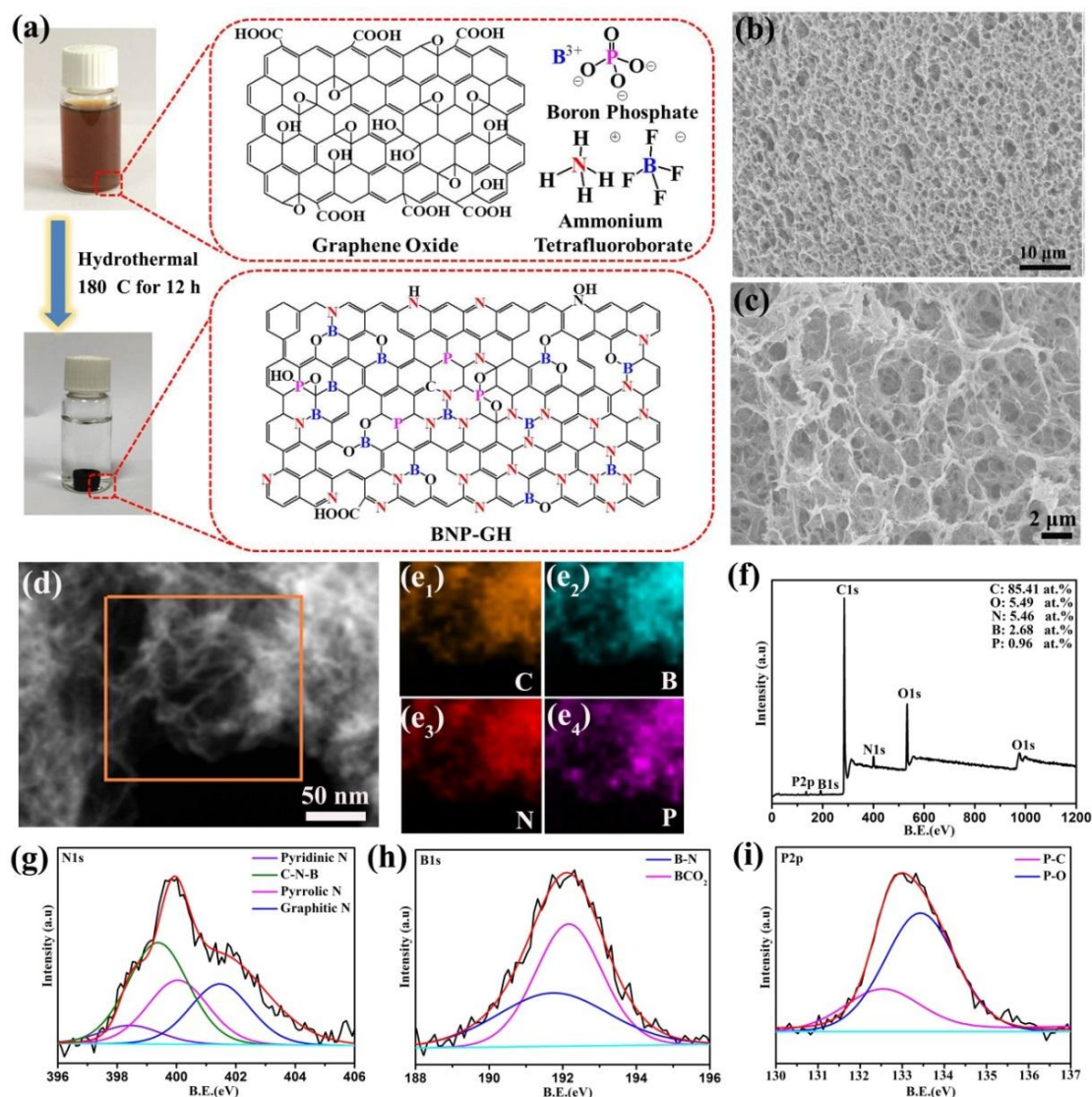


Figure 1. (a) Schematic illustration of the preparation of BNP-GH. (b) Low- and (c) high-magnification SEM images of the interior microstructure of the freeze-dried BNP-GH. (d) STEM image and (e1) carbon, (e2) boron, (e3) nitrogen, and (e4) phosphorous element mapping images of BNP-GH taken in the square region from Figure 1d. XPS (f) survey, (g) B1s, (h) N1s and (i) P2p spectra of BNP-GH.

The synthetic strategy of boron, nitrogen, and phosphorus ternary-doped graphene hydrogel (BNP-HGH) is schematically illustrated in Figure 1a. In a brief, an aqueous dispersion of GO (2 mg mL^{-1}) containing boron phosphate (BPO_4) and ammonium tetrafluoroborate (NH_4BF_4) was hydrothermally treated at 180 °C for 12 h

to obtain BNP-GH. More synthesis detail can be found in the experimental section. Moreover, the boron and nitrogen co-doped GH (BN-GH), nitrogen and phosphorus co-doped GH (NP-GH), nitrogen doped GH (N-GH) and pure GH were also prepared as control samples using the same procedure with the addition of corresponding dopants as needs (see experimental section). Noted the size and shape of BNP-GH can be easily adjusted by changing the type of reactors (Figure S2). The microstructure of this BNP-GH was characterized by scanning electron microscopy (SEM). As shown in Figure, 1b and 1c, an interconnected 3D porous framework with the pore size ranging from a few hundred nanometers to several micrometers can be discerned (Figure S3 and S4). Similar morphology can be observed for BN-GH, NP-GH, N-GH and GH control samples (Figure S5). The porous structure of the BNP-GH was further investigated by nitrogen adsorption-desorption measurements (Figure S6). The isotherm curve exhibits a type-IV behavior with pronounced adsorption, indicating the existence of a large number of mesopores and macropores in the 3D network of BNP-GH. Brunauer–Emmett–Teller (BET) and Barrett–Joyner–Halenda (BJH) analysis also reveal that the BNP-GH has a high specific surface area of $\sim 498 \text{ m}^2 \text{ g}^{-1}$, pore volume of $0.85 \text{ cm}^3 \text{ g}^{-1}$ and pore size of 2-65 nm.

The dispersion of doping elements in the BNP-GH was unraveled by the elemental mapping analysis. As shown in Figure 1d and 1e, B, N and P elements are homogeneously doped into the porous graphene sheets. X-ray photoelectron spectroscopy (XPS) is used to characterize the doping content and the valence states of the individual elements in BNP-GH. The typical survey C1s, N1s, B1s and P2p

spectra for BNP-GH were illustrated in Figure 1f. The quantitative XPS analysis exhibits a doping content of 5.46 at % N, 2.68 at % B and 0.96 at % P in BNP-GH, which is similar to the results analyzed by energy-dispersive X-ray (EDX) spectra (S7).^{22, 29} In Figure 1g, the high resolution N1s XPS spectrum of BNP-GH can be deconvoluted into four different signals with binding energies of 398.4, 399.4, 400.1 and 401.5 eV, corresponding to pyridinic N, C-N-B, pyrrolic N, and graphitic N, respectively. Two peaks of B1s XPS spectra in Figure 1h can be identified to B-N and BCO₂ (Table S2).⁸ In the P2p XPS spectrum (Figure 1i), two distinguishable peaks at 132.7 eV and 133.8 eV correspond to P-C and P-O, respectively.³³⁻³⁴ All these results well demonstrate that N, B and P elements are successfully doped into the carbon framework of BNP-GH. Raman spectra further provide additional evidence of the introduction of N, B and P atoms into graphene layer of BNP-GH. The ratio of the intensity of D band to G band (I_D/I_G) is normally used to characterize the disorder of carbon materials.³⁵⁻³⁶ As shown in Figure S8, it can be found that the doping GH has a relatively larger I_D/I_G , i.e., BNP-GH (1.21), BN-GH (1.14), NP-GH (1.15) and N-GH (1.03), compared with non-doping GH (0.95), which is caused by the rich C-B, C-P, and C-N bonds.

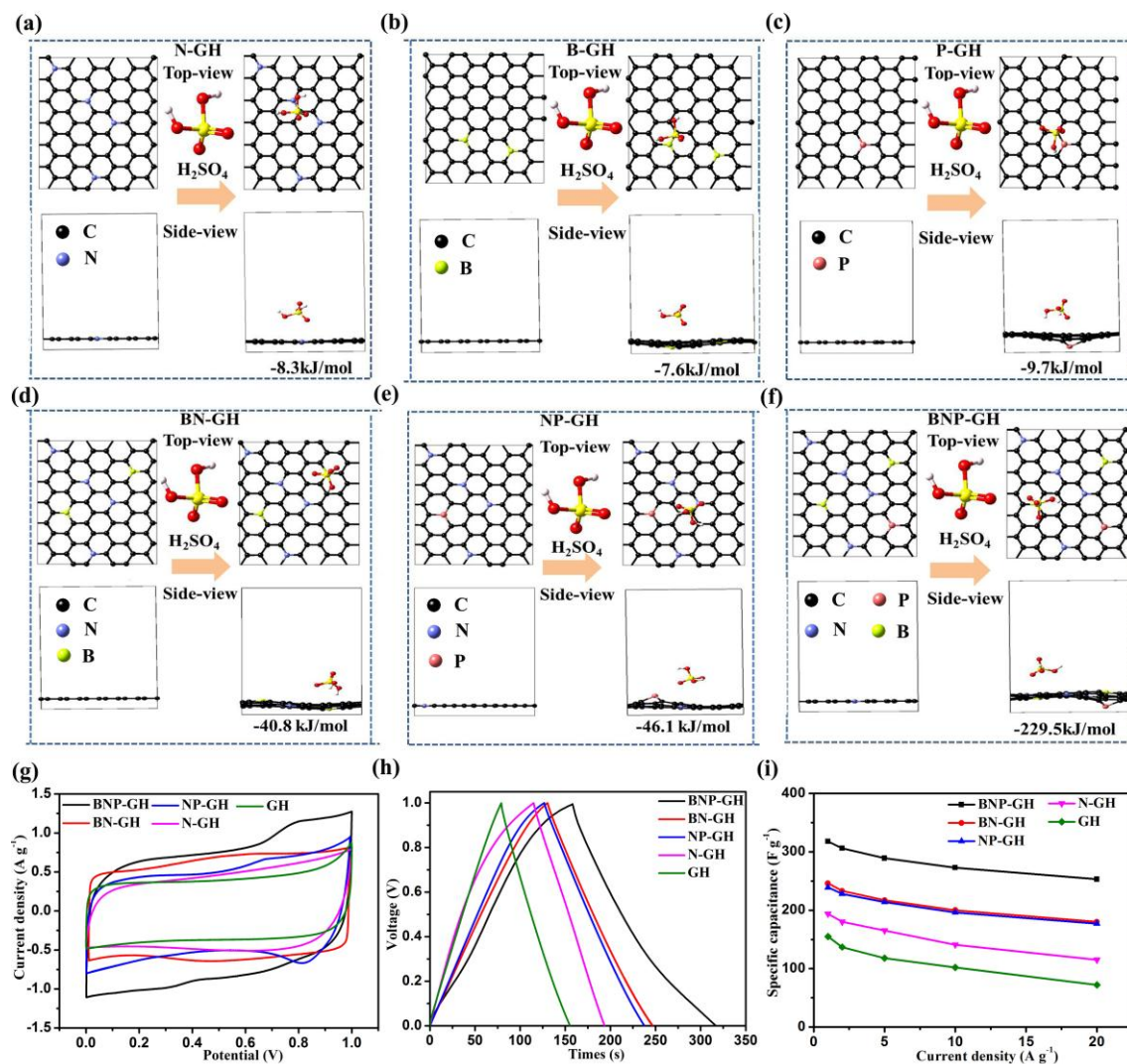


Figure 2. The ab initio calculated structures and energies of H_2SO_4 adsorption on the N doped GH (N-GH) (a); B doped GH (B-GH) (b); P doped GH (P-GH) (c); B, N codoped GH (BN-GH) (d); N, P codoped GH (NP-GH) (e), and B, N, P ternary doped GH (BNP-GH) (f). (g) Cyclic voltammetry (CV) curves of 5 mV s^{-1} and (h) galvanostatic charge-discharge (GCD) curves of 1 A g^{-1} of BNP-GH, BN-GH, NP-GH, N-GH and GH based symmetric SCs. (i) Comparison of specific capacitances versus different current densities.

To investigate the influence of these doped heteroatoms on the electrochemical behavior of BNP-GH films, first-principle calculations using density functional theory

(DFT) were performed. The calculated structures of one H_2SO_4 molecule adsorbed on GH, N-GH, B-GH, P-GH, BN-GH, NP-GH and BNP-GH were shown in Figure 2(a-f) and Figure S9. It can be found that the adsorption energy of ternary-doped BNP-GH with H_2SO_4 ($229.5 \text{ kJ mol}^{-1}$) is about 5.5 times higher than that of co-doped GHs such as BN-GH (40.8 kJ mol^{-1}) and NP-GH (46.1 kJ mol^{-1}), about 27 times higher than that of mono-doped GH, such as B-GH (7.6 kJ mol^{-1}), N-GH (8.3 kJ mol^{-1}) and P-GH (9.7 kJ mol^{-1}) and about 34 times higher than that of un-doped GH (6.8 kJ mol^{-1}). The enhancement on the adsorption energy for BNP-GH demonstrates that ternary doping with B, N and P elements has a synergistic effect in improving the compatibility of BNP-GH with H_2SO_4 electrolyte and leading to a superior electrochemical performance. In addition, we have also explored such effect in the EMIMBF₄ electrolyte system. Similarly, the adsorption energy of BNP-GH with EMIMBF₄ is the highest, as shown in Figure S10. These results well demonstrate that the ternary doped graphene electrode is compatible not only with aqueous electrolyte, but also with organic electrolyte, suggesting that our BNP-GH electrode is a promising electrode in various electrolyte systems.

To verify these calculation results, we studied the electrochemical performances of the symmetric SCs based on BNP-GH, BN-GH, NP-GH, N-GH and GH as electrodes. Hydrogel monoliths were directly fabricated as binder-free electrodes by cutting and pressing the GHs into thin round films (Figure S11a and the details in experimental section). Interestingly, although the doped graphene framework became flat and crumpled upon physical pressing, the 3D continuous porous network in

microstructure can be well maintained, which will be beneficial for the electrolyte infiltration and ions diffusion (Figure S11). Figure 2g and Figure S12a show the cyclic voltammetry (CV) curves of different GH electrodes based SCs at a scan rate from 5 to 50 mV s^{-1} . It can be seen that all doped GHs exhibit a large capacitive response, accompanied by weak redox peaks, indicative of the presence of a pseudocapacitive effect with heteroatom doping.¹² Among all those doped GH electrodes, BNP-GH shows the largest capacity (Figure 2h), suggesting that B, N and P have strong synergistic doping effects, which is in good agreement with the calculation results. As a result, the BNP-GH delivers an impressive specific capacitance of 318 F g^{-1} at a current density of 1 A g^{-1} (Figure 2i and Figure S12), more than 2.0 times that of GH (152 F g^{-1}) (~ 1.3 times that of BN-GH, NP-GH and ~ 1.6 times that of N-GH).

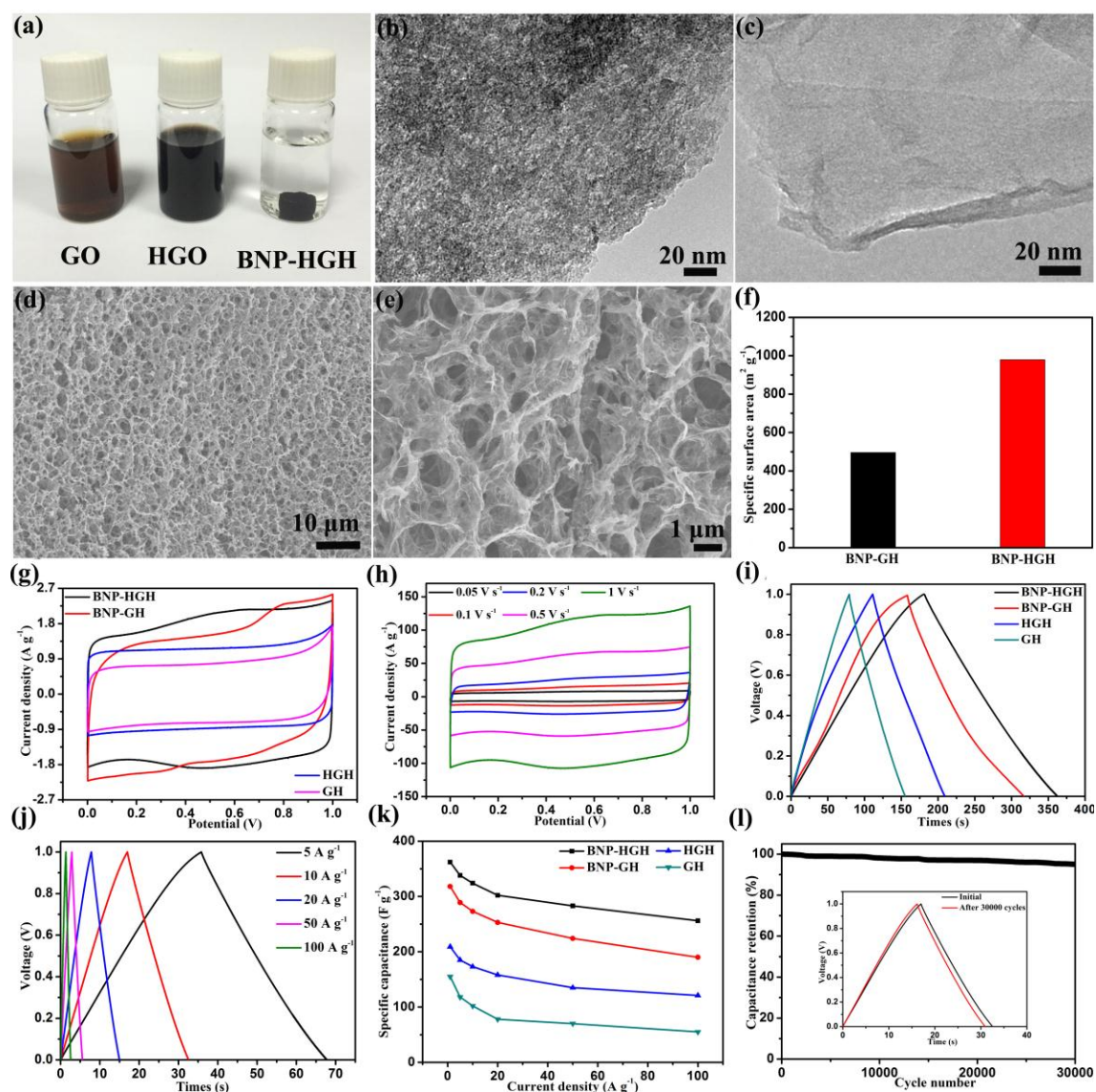


Figure 3. (a) Photographs of aqueous dispersions of GO, HGO and as-prepared BNP-HGH. (b), (c) TEM images of HGO and GO. (d) Low- and (e) high-magnification SEM images of BNP-HGH. (f) Specific surface areas of BNP-GH and BNP-HGH. (g) Cyclic voltammetry (CV) curve of 10 mV s⁻¹ of GH, HGH, BNP-GH, and BNP-HGH based symmetric SCs. (h) CV curves of BNP-HGH based symmetric SC at different scan rates. (i) Galvanostatic charge-discharge (GCD) curves of 1 A g⁻¹ of GH, HGH, BNP-GH, and BNP-HGH based symmetric SCs. (j) GCD curves of BNP-HGH based symmetric SC at different current densities. (k) Comparison of specific capacitances

versus different current densities. (l) Cycling stability of the BNP-HGH based SC at a current density of 10 A g^{-1} . The inset of (l) shows the charge-discharge curves for the 1st and 30000th cycle.

With large ion-accessible surface area and efficient ion transport pathways, the holey graphene based electrodes have shown great advantages in SCs.³⁷ Therefore, in order to further improve the performance of BNP-GH based SCs, we employed holey graphene oxide (HGO) with abundant in-plane nanopores obtained by a mild defect-etching reaction²⁴ (Figure S13) instead of previous GO to prepare BNP-HGH according to the same method for BNP-GH (Figure 3a). High-resolution transmission electron microscopy (HR-TEM) reveals that abundant in-plane pores with a few nanometers (2-5 nm) across the entire plane of HGO sheets (Figure 3b), in clear contrast to nanopore-free GO sheets (Figure 3c), suggesting an efficient etching of carbon atoms in GO by H_2O_2 . Similar to the morphology of BNP-GH, the freeze-dried BNP-HGH also shows an interconnected 3D macroporous network and the pore walls consisting of thin layers of graphene sheets (Figure 3d and 3e). In addition, as shown in Figure S14, the BNP-HGH still retains the rich heteroatom-doped elements. It is worth noting that the freeze-dried BNP-HGH has a BET specific surface area of $\sim 980 \text{ m}^2 \text{ g}^{-1}$ (Figure 3f and Figure S15), about 2.0 times higher than that of BNP-GH ($\sim 498 \text{ m}^2 \text{ g}^{-1}$), indicating that the enriched in-plane nanopores not only offer a short pathway for fast ionic diffusion across stacked graphene layers, but also significantly increase the accessible surface area.

Because of above mentioned merits, the CV (Figure 3g) and galvanostatic

charge-discharge (GCD) studies (Figure 3j) show a greatly enhanced electrochemical performance for BNP-HGH based SCs in comparison with BNP-GH. The nearly rectangular CV curves at a high scan rate of 1 V s^{-1} (Figure 3h) and the nearly symmetric triangle GCD curves at a high current density of 100 A g^{-1} (Figure 3i) indicate a nearly ideal EDLC behavior and efficient electrolyte ion transport throughout the BNP-HGH electrodes. The calculated specific capacitances at different current densities from 1 A g^{-1} to 20 A g^{-1} are shown in Figure 3k. Notably, the BNP-HGH electrode exhibits an ultrahigh specific capacitance of 362 F g^{-1} at a current density of 1 A g^{-1} , 14 % higher than that of BNP-GH (318 F g^{-1}) and also superiors to most reported values (Table S3). Impressively, the BNP-HGH can retain as high as $\sim 71 \%$ of its initial value (256 F g^{-1}) when increasing the current density to 100 A g^{-1} ; while the BNP-GH only shows a 60 % capacitance retention (190 F g^{-1}). BNP-HGH based SC also exhibits a great cycle stability, i.e., 95.1 % of the initial capacitance retention after 30000 cycles at a current density of 10 A g^{-1} , compared to the values of 89.3 % for BNP-GH, 82.5 % for HGH and 73.8 % for GH (Figure 3l and Figure S16). The minimal change in the discharge-charge curves before and after cycling 30000 cycles (the inset in Figure 3l) again suggests the excellent electrochemical reproducibility and long term cycle stability of our BNP-HGH based SC.

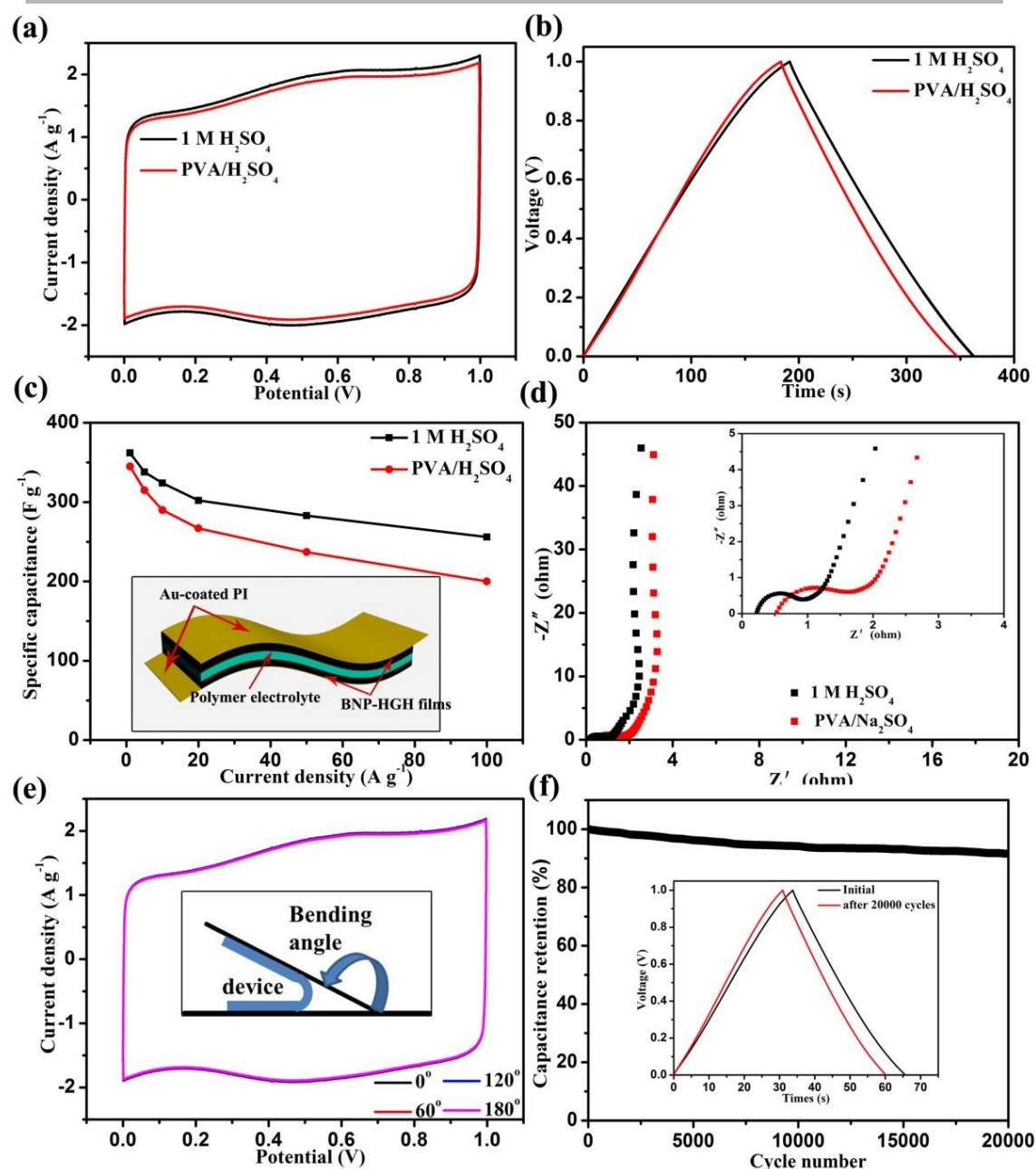


Figure 4. Comparison of (a) CV curves at 10 mV s⁻¹ and (b) GCD curves at 1 A g⁻¹ of BNP-HGH based all-solid-state supercapacitors (ASCs) in 1 M H₂SO₄ aqueous electrolyte and in 1 M H₂SO₄/PVA gel electrolyte. Comparison of (c) specific capacitances and (d) Nyquist plots of BNP-HGH based ASCs in 1 M H₂SO₄ aqueous electrolyte and in H₂SO₄/PVA gel electrolyte. The inset in (d) shows the magnified high-frequency regions of the Nyquist curves. (e) CV curves of the ASC at 10 mV s⁻¹ for different bending angles. (f) Cycling stability of the ASC at a current density of 10

A g^{-1} under bending state. The inset of (f) shows the charge-discharge curves for the 1st and 20000th cycle.

In order to demonstrate the remarkable performances of BNP-HGH electrodes for electrochemical energy storage in flexible and wearable electronics, we have also assembled the flexible all-solid-state supercapacitors (ASCs) based on BNP-HGH in 1 M H_2SO_4 /polyvinyl alcohol (PVA) gel electrolyte. BNP-HGH can function as an ideal electrode for ASC devices due to the exceptional mechanical and electrical robustness of the highly interconnected 3D network. The BNP-HGH based ASC shows almost the same CV curve with the one in 1 M H_2SO_4 aqueous electrolyte, indicating an efficient infiltration of H_2SO_4 /PVA gel electrolyte into the 3D network of BNP-HGH (Figure 4a). The accurate specific capacitances are calculated from GCD measurements (Figure 4b and Figure S17). At a current density of 1 A g^{-1} , the BNP-HGH electrode in gel electrolyte delivers a comparable specific capacitance (345 F g^{-1}) to the one in aqueous electrolyte (362 F g^{-1}), which is consistent with the CV analysis (Figure 4c). Increasing the current density to 100 A g^{-1} , it can be found that the specific capacitance of BNP-HGH in gel electrolyte reduces to 200 F g^{-1} (58 % of that at 1 A g^{-1}), 25 % lower than the one in aqueous electrolyte (256 F g^{-1} , 70 % of that at 1 A g^{-1}). This can be ascribed to the increased viscosity of gel electrolyte compared to the aqueous electrolyte, and thus lead to a higher internal resistance and slower ion diffusion rate in the ASC. Indeed, Nyquist plots in Figure 4d confirm that the internal resistance of the ASC in gel electrolyte is higher than that in aqueous electrolyte. Nonetheless, our BNP-HGH based ASC exhibits a superior performance comparing to

other all-solid-state devices, i.e. its specific capacitance is substantially higher than those made by carbon nanotubes, graphene films and their composites (Table S4).

Our flexible ASC also exhibits excellent mechanical robustness in the bending test. The CV curves of the ASC measured at various bending angles show nearly the same capacitive behavior, even at a large bending angle of 180° (Figure 4e), indicating the great electrochemical performance retention of our device under tough working environment. The cycling stability of this ASC was further characterized by GCD tests up to 20000 cycles under 180° bending angle at a high current density of 10 A g^{-1} (Figure 4f). Remarkably, only 8.4 % decay in specific capacitance is observed, highlighting the exceptional mechanical and electrical robustness of this BNP-HGH electrode benefiting from its highly interconnected 3D network and the favorable interfacial electrochemical behaviors of the BNP-HGHs in the $\text{H}_2\text{SO}_4/\text{PVA}$ gel electrolyte.

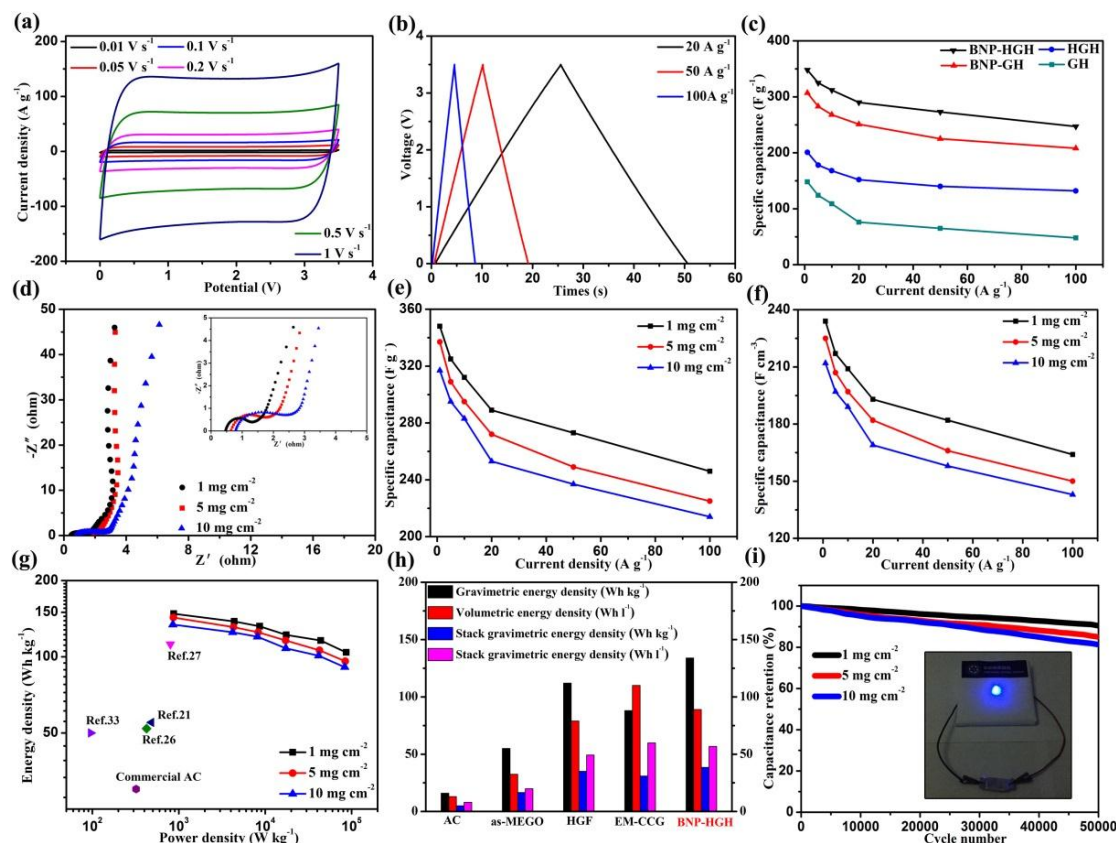


Figure 5. CV curves (a) and GCD curves (b) of the BNP-HGHs based SC in 1 M EMIMBF₄/AN electrolyte. (c) Comparison of specific capacitances of GH, HGH, BNP-GH, and BNP-HGH electrodes at different current densities. (d) Nyquist plots of the BNP-HGHs based SCs (with graphene mass loading of 1, 5 and 10 mg cm⁻²) and the inset shows the magnified high-frequency regions of the Nyquist curves. Gravimetric (e) and volumetric (f) specific capacitances of the BNP-HGH electrodes at different current densities. (g) Ragone plots of gravimetric energy density versus power density for the BNP-HGHs based SCs in comparison with some other SCs reported in literature (using electrodes with the graphene mass loading of ~ 10 mg cm⁻²). (h) Gravimetric and volumetric energy densities (normalized by the total weight or volume of both electrodes in each SC), and stack gravimetric and volumetric energy densities (normalized by the total weight or volume of the whole SC including two

electrodes, two current collectors, electrolyte, separator, and the wrapping parafilm package) for SCs made from various carbon materials (using electrodes with the graphene mass loading of $\sim 10 \text{ mg cm}^{-2}$). (i) Cycling stability of BNP-HGH based SCs at a current density of 10 A g^{-1} (with graphene mass loading of 1, 5 and 10 mg cm^{-2}).

As we all know, the narrow voltage window for aqueous electrolytes is a critical limiting factor for high energy SCs.^{9,14} A higher operation voltage corresponds to a higher overall performance of the SC, as the energy density and power density are both proportional to the square of the voltage ($E = 1/2 CV^2$ and $P_{\text{max}} = V^2/4R$, where C is the cell capacitance, V is the voltage and R is the ESR). Herein, we have also evaluated the electrochemical performances of the BNP-HGHs based SC in an ionic liquid electrolyte (1.0 M 1-ethyl-3-methylimidazolium tetrafluoroborate (EMIMBF₄) in acetonitrile (AN). In contrast to the electrochemical behavior in 1 M H₂SO₄ aqueous electrolyte, no apparent redox peaks of the BNP-HGHs based SC are observed in the 1 M EMIMBF₄/AN ionic liquid electrolyte. As shown in Figure 5a, all CV curves are close to rectangular shape at 0–3.5 V even under a high scan rate of 1000 mV s^{-1} . Consistently, GCD curve shows a nearly symmetric triangular shape at the current density of 100 A g^{-1} with a small voltage (IR) drop of 0.32 V (Figure 5b), indicating the excellent rate performance of this BNP-HGH electrode. At room temperature, the SC based on BNP-HGH electrode in ionic liquid electrolyte delivers a specific capacitance of up to 350 F g^{-1} at a current density of 1 A g^{-1} (Figure 5c), only 3.3 % lower than that in aqueous electrolyte (362 F g^{-1}). Even at 100 A g^{-1} , the

BNP-HGH electrodes still retain a gravimetric capacitance of 202 F g^{-1} . The achievement of such high gravimetric specific capacitance in organic electrolyte (350 F g^{-1}) leads to an ultrahigh gravimetric energy density of 148 Wh kg^{-1} (Table 5), which is almost comparable to that of lithium ion batteries ($\approx 150 \text{ Wh kg}^{-1}$, against the mass of the active materials in the electrodes) (Figure 5g). In addition, on the basis of their packing density of 0.67 g cm^{-3} , the volumetric capacitance and energy density of BNP-HGH electrodes were calculated to be 234 F cm^{-3} and 99 Wh L^{-1} , respectively, also the best values achieved in all carbon-based SC electrodes to date. (Figure 5f, S19 and Table S5).

To further evaluate the performance of our BNP-HGH electrodes for practical application, we have also investigated the electrochemical properties of BNP-HGH electrodes with a graphene mass loading of $\sim 10 \text{ mg cm}^{-2}$ (electrode thickness of $150 \mu\text{m}$). Impressively, no sharp decay on specific capacitance is observed when the graphene mass loading dramatically increases from 1 (350 F g^{-1}) to 10 mg cm^{-2} (316 F g^{-1}) at a current density of 1 A g^{-1} (Figure 5e, Figure S18 and Table S6). The excellent performance is benefited from the high electrical conductivity and fast ion diffusion of the BNP-HGH electrode, which can be reflected by the lower resistance shown in Figure 5d. As a consequence, a gravimetric energy density of 134 Wh kg^{-1} at a gravimetric power density of 83 kW kg^{-1} is achieved with a practical graphene mass loading level (Figure 5g). Accordingly, the volumetric capacitance and volumetric energy density of BNP-HGH electrode slightly decrease to 212 F cm^{-3} and 89 Wh L^{-1} (Figure 5f and Figure S19), respectively. It is worth noting that both the gravimetric

and volumetric energy density are comparable to or better than those of the best carbon-based nanomaterials including activated carbon,⁵ carbon nanotubes¹⁰ and graphene⁹ with large areal mass loadings reported to date (Table S5). In addition, the gravimetric and volumetric energy density against the whole SC stack — including two electrodes, two current collectors, electrolyte, one separator and packaging — are further studied to evaluate the real potential applications of our BNP-HGH electrode as high energy SCs (the details see experimental section). The BNP-HGH electrodes contribute 28.7 % weight and 64.5 % volume for the whole device, and thus it can be calculated that our device delivers a stack gravimetric and volumetric energy densities of 38.5 Wh kg⁻¹ and 57.4 Wh L⁻¹, respectively (Figure 5h and Table S5-6), which are nearly one order of magnitude higher than those of commercial AC based SCs (4–5 Wh kg⁻¹ and 5–8 Wh L⁻¹) and comparable to those of lead-acid batteries (35–40 Wh kg⁻¹ and 50–90 Wh L⁻¹).

The cycle life of an SC is another important parameter to evaluate its capability for practical application. Our BNP-HGH based SC in EMIMBF₄/AN electrolyte shows superior cycling stability with a ~ 90.5% capacitance retention after 50,000 charge-discharge cycles at a high current density of 20 Ag⁻¹ (85.1 % for 5 mg cm⁻² and 81.3 % for 10 mg cm⁻², Figure 5i), again demonstrating its promising potential for the practical application. After charging, the BNP-HGH based SC can light up a blue light emitting diode (LED, 3.0-3.2 V) indicator about 60 s (inset of Figure 5(i) and Video 1).

In commercial SCs, AN was commonly used as electrolyte solvent because of its

high ionic conductivity (about 60 S m^{-1}),^{4,15} however, this kind of AN-based electrolyte is not suitable for high-temperature operation due to its low boiling temperature. In contrast, the EMIMBF₄ electrolyte can work well at high temperature. In this work, the electrochemical performance of the BNP-HGH electrode in the pure EMIMBF₄ electrolyte is also tested, and the measured results show the nearly same specific capacitance (340 F g^{-1} and 228 F cm^{-3}) and energy densities (144 Wh kg^{-1} and 96.8 Wh L^{-1}) (Figure S20). The achieving high capacitance without AN, further demonstrates the unique advantage of this BNP-HGH electrode for efficient electrolyte transport and surface area access, comparing with previous studies.^{21, 27}

4. Conclusions

In summary, a binder-free BNP-HGH electrode was synthesized via a facile two step method: (1) solution processable HGO with abundant in-plane nanopores via a convenient mild defect-etching reaction, and (2) HGO based hydrothermal process in presence of BPO₄ and NH₄BF₄ as precursors. With high electrical conductivity (Table S7), large ion accessible surface area and efficient ion transport pathway, the BNP-HGH electrode exhibited outstanding electrochemical performances, including the highest gravimetric capacitance of 350 F g^{-1} and volumetric capacitance of 234 F cm^{-3} to date for graphene-based SCs in an ionic liquid electrolyte. Importantly, the high graphene mass loading BNP-HGH electrode ($\sim 10 \text{ mg cm}^{-2}$) based SCs achieved high stack gravimetric and volumetric energy densities that are comparable to those of lead-acid batteries, but with much higher power density. The achievement of such high energy density in BNP-HGH based SCs bridges the gap between traditional

capacitors and batteries, and can open up exciting opportunities for mobile power supply in a wide range of applications, including electrical vehicles and mobile electronics.

Competing financial interests

The authors declare no competing financial interests.

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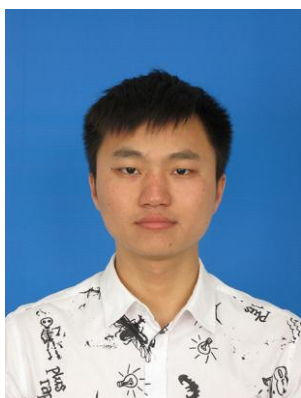
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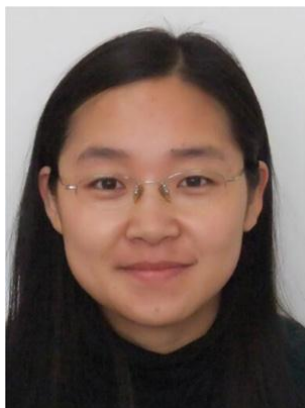


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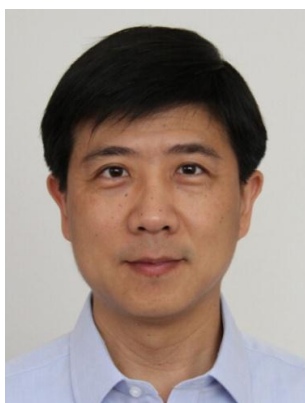


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Highlights

- A three-dimensional boron, nitrogen, and phosphorus ternary doped holey graphene hydrogel (BNP-HGH) film with high electrical conductivity, large ion accessible surface area, efficient electron and ion transport pathways as well as high ion adsorption capacity is prepared by a facile method.
- The BNP-HGH film is directly used as a binder-free supercapacitor (SC) electrode and delivers the highest specific capacitance of 350 F g^{-1} (234 F cm^{-3}) for graphene-based SCs using an organic electrolyte to date.
- With a commercial level graphene mass loading ($\sim 150 \text{ }\mu\text{m}$, $\sim 10 \text{ mg cm}^{-2}$), the BNP-HGH electrode exhibits ultrahigh gravimetric and volumetric energy densities of 134 Wh kg^{-1} , and 89 Wh L^{-1} , respectively.
- Normalized by the total weight or volume of the whole SC, the device still shows an impressive stack gravimetric and volumetric energy densities of 38.5 Wh kg^{-1} and 57.4 Wh L^{-1} , respectively, comparable to those of lead-acid batteries ($35\text{--}40 \text{ Wh kg}^{-1}$ and $50\text{--}90 \text{ Wh L}^{-1}$).